M1R Random Matrices

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Contents

In these notes we introduce random matrices, introducing some basic theory combined with practical experiments. Much of the theory goes beyond the knowledge of 1st year students so these notes are really just the tip-of-the-iceberg: random matrices have a very rich deeper theory combining aspects of probability, statistics, combinatorics, group theory, approximation theory, functional analysis, complex analysis, integrable systems, statistical mechanics, and numerical analysis.

We divide the notes into three chapters:

I. *Universality.* Eigenvalue statistics have universality properties that emerge independent of the details of distribution of their entries. We introduce this phenomena by an analogy with the Central Limit Theorem (CLT, which we also introduce), and explore both global and local statistics.

II. *Free Probability.* Algebraic manipulations of random matrices have the property that their eigenvalue distribution can be deduced via *Free Probability*, which can be viewed as a non-commutative version of probability that originally arose in group theory. This leads to a free version of convolutions. We observe this phenomena experimentally and argue why such a convolution exists.

III. *Randomised Linear Algebra.* We can incorporate randomness into linear problems like least squares (i.e., regression) and achieve fairly accurate results with much fewer computational cost. This chapter introduces least squares and explores, experimentally, the practical realisation of randomised linear algebra.

Chapter I

Universality

In this chapter we introduce the concept of *universality*: eigenvalue statistics of random matrices are largely independent of the distribution of the entries. In particular, we will look at both *global statistics*, i.e., what do histograms of all eigenvalues look like, and *local statistics*, i.e., what do histograms of the largest eigenvalue look like. For the theory we focus on the global statistics (as proving results for the local statistics requires much more sophisticated mathematical tools), and in particular, we focus on the phenomenon that symmetric matrices with independent, identically distributed (iid) entries have a global statistics that look like the *semicircle*:

$$\frac{\sqrt{4-x^2}}{2\pi} \qquad \text{for } x \in [-2,2] \text{ and } 0 \text{ otherwise.}$$

The *semicircle distribution* can be viewed as a random matrix analogue of the *normal distribution* $\mathcal{N}$, whose Probability Density Function (PDF) is a Gaussian:

$$\frac{\mathrm{e}^{-x^2/2}}{\sqrt{2\pi}}.$$

In this section we will show universality using the *Method of Moments*: we will show that the moments of the statistics converge to the moments of a known distribution. For a scalar random variable X the distribution is generally encoded by the standard moments:

$$\mathbb{E}X^k$$

We will see that in the matrix case the global distribution of eigenvalues is encoded by the matrix-analogue of moments:

$$\mathbf{E}A^k := \mathbb{E}\mathrm{Tr}\, A^k$$

where Tr is the trace of a matrix (sum of the diagonal).

In particular, we will first discuss the Central Limit Theorem (CLT), a scalar version of universality where the moments of rescaled sums of random variables with finite moments always tend to the moments of a Gaussian. We will then see a similar result for semicircle law, where rescaled matrix moments tend to the moments of a semicircle as the matrix size tends to infinity.

Note that moments are only guaranteed to uniquely define a density in the special case where the probability density decays sufficiently fast, but this is something the Gaussian and semicircle both satisfy.

I.1 Experimental observation of universality

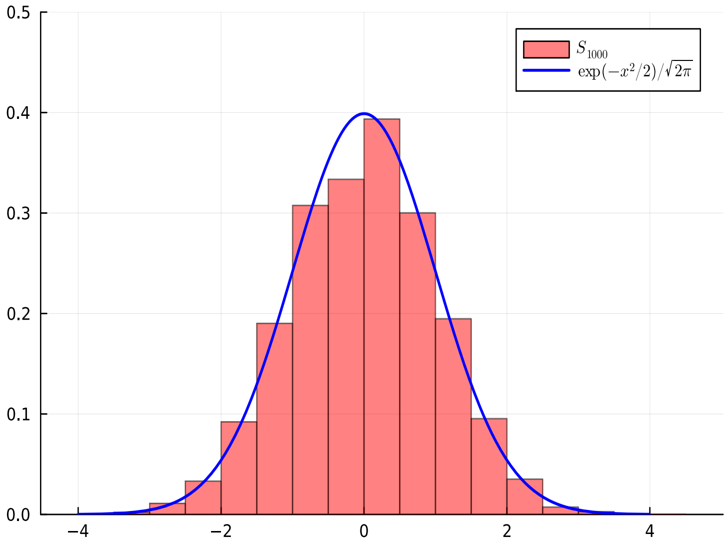
Before digging into the theory we will do an experimental exploration of universality. Let's begin with the classical Central Limit Theorem (CLT). Consider sampling iid discrete random variables $X_1, \ldots, X_n$ each taking values ± 1 with equal probability, so that their expectation is zero and variance is one:

$$\mathbb{E}X_k = (-1)\frac{1}{2} + 1\frac{1}{2} = 0, \qquad \mathbb{E}X_k^2 = (-1)^2\frac{1}{2} + 1^2\frac{1}{2} = 1.$$

By summing and rescaling these variables we can get a new random variable S_n:

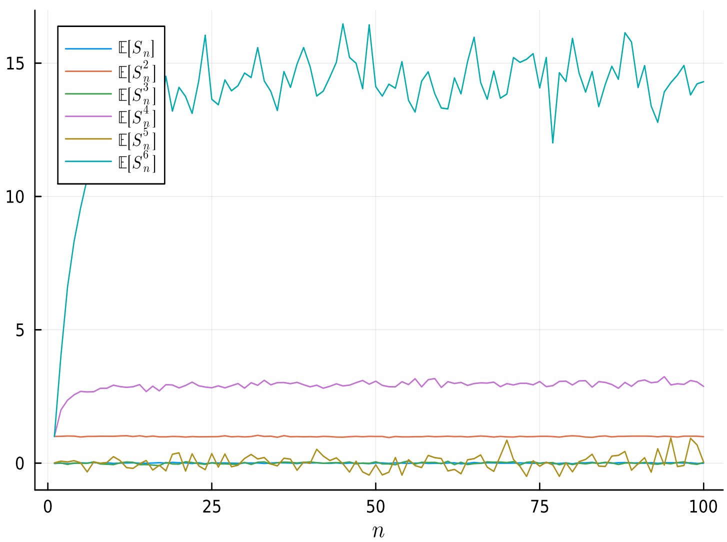
$$S_n := \frac{X_1 + \cdots + X_n}{\sqrt{n}}.$$

For large n we can compare a histogram of samples of S_n to a Gaussian:



This phenomena is independent of the distribution of X_k provided the moments are finite: we always get a normal distribution (shifted and scaled to match the mean and variance).

We are going to show why this phenomena is true using the Method of Moments: we want to show the moment $\mathbb{E}[S_n^k]$ tend to the moments of a Gaussian. We can deduce the moments experimentally by averaging over many samples and letting n increase:

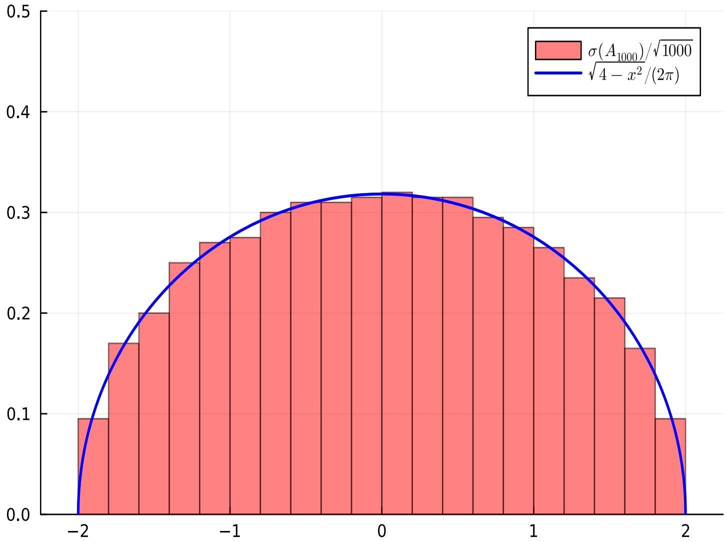


A sensible conjecture then is that $\mathbb{E}[S_n^k]$ is zero when k is odd and

$$\mathbb{E}[S_n^2] \to 1, \quad \mathbb{E}[S_n^4] \to 3, \quad \mathbb{E}[S_n^6] \to 15.$$

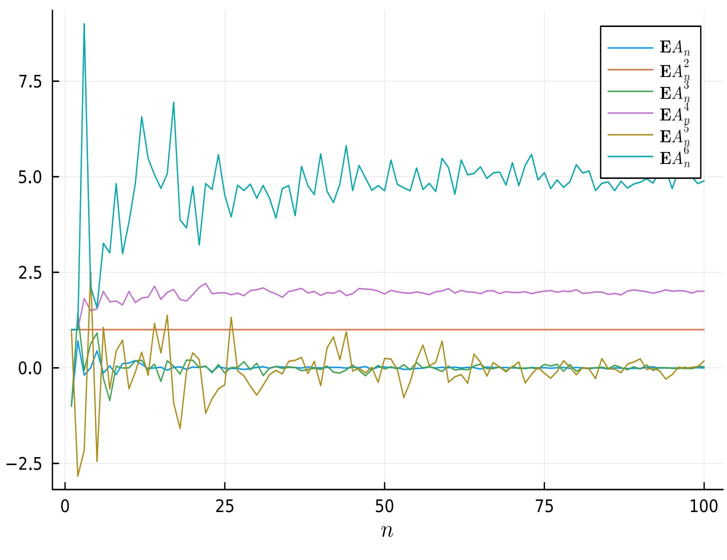
Thus we are left with the tasks of: (1) deducing an exact formula for these moments, and (2) proving that the Gaussian has the same moments.

The matrix version of universality involves a histogram of the eigenvalues. Consider $A_n \in \mathbb{R}^{n\times n}$ which is symmetric ($A_n = A_n^\top$) and whose entries on and above the diagonal are iid random variables taking values ± 1 with equal probability (and entries below the diagonal defined by symmetry). Each sample of A_n will have n eigenvalues, which we denote $\sigma(A_n) = \{\lambda_1, \ldots, \lambda_n\}$ (for concreteness we can assume they are sorted from smallest to largest). If we plot a histogram of rescaled eigenvalues $\frac{\sigma(A_n)}{\sqrt{n}} = \left\{\frac{\lambda_1}{\sqrt{n}}, \ldots, \frac{\lambda_n}{\sqrt{n}}\right\}$ of a *single sample* A_n we get a semicircle:



Just as in the CLT, this phenomena is independent of the distribution of the entries of A_n provided the moments are finite: we always get a semicircle (shifted and scaled to match the mean and variance).

Like the CLT we will show this observation using the Method of Moments, and again we can deduce the moments experimentally by letting n increase (a single sample per n works in this case):



Again the odd moments appear to be zero but now we might conjecture that the even moments satisfy

$$\mathbf{E}A_n^2 \to 1, \quad \mathbf{E}A_n^4 \to 2, \quad \mathbf{E}A_n^6 \to 5.$$

Again we are left with the tasks of: (1) deducing an exact formula for these moments, and (2) proving that the semicircle has the same moments.

In the remainder of this chapter we sketch out a proof of both the CLT and the semicircle law for random matrices.

I.2 Moments of the normal distribution

We will use the method of moments, so our first step is to establish an explicit formula for the moments of a Gaussian. We will think of this in terms the following combinatorial object:

Definition 1 (double factorial)**.**

$$(2m-1)!! := \frac{(2m)!}{2^m m!} = 1 \cdot 3 \cdot 5 \cdots (2m-1).$$

Note this satisfies a simple recurrence:

$$(2m-1)!! = (2m-1) \cdot (2m-3)!! \quad \text{for} \quad m > 1.$$

We can check the first few double factorials:

$$1!! = 1, \quad 3!! = 3, \quad 4!! = 15$$

which exactly match our predicted values for the even moments $\mathbb{E}[S_n^k]$.

To prove this result we will use a combinatorial argument. Note that the double factorial counts the number of ways to partition $\{1, \ldots, 2m\}$ into m different pairs, that is to write

$$\{1, \ldots, 2m\} = \{x_1, y_1\} \cup \cdots \cup \{x_m, y_m\}$$

where $x_1, \ldots, x_m, y_1, \ldots, y_m$ is a permutation of $1, \ldots, 2m$ and $x_k < y_k$ for all k. For example, if $m = 2$ then we have the following pairings of $1, 2, 3, 4$:

$$\{1,2\},\{3,4\} \qquad \{1,3\},\{2,4\} \qquad \{1,4\},\{2,3\}$$

and thus $3!! = 3$. And if $m = 3$ then we have $5!! = 15$ total pairings:

$$\begin{array}{lll}
\{1,2\},\{3,4\},\{5,6\} & \{1,2\},\{3,5\},\{4,6\} & \{1,2\},\{3,6\},\{4,5\} \\
\{1,3\},\{2,4\},\{5,6\} & \{1,3\},\{2,5\},\{4,6\} & \{1,3\},\{2,6\},\{4,5\} \\
\{1,4\},\{2,3\},\{5,6\} & \{1,4\},\{2,5\},\{3,6\} & \{1,4\},\{2,6\},\{3,5\} \\
\{1,5\},\{2,3\},\{4,6\} & \{1,5\},\{2,4\},\{3,6\} & \{1,5\},\{2,6\},\{3,4\} \\
\{1,6\},\{2,3\},\{4,5\} & \{1,6\},\{2,4\},\{3,5\} & \{1,6\},\{2,5\},\{3,4\}.
\end{array}$$

This can be deduced by a very simple inductive argument. When $m = 1$ we have exactly $1!! = 1$ pairings. For $m > 1$, assuming the number of pairings for $n < m$ is given by $(2n-1)!!$. We can pair 1 with a choice of the $2m-1$ numbers: $2, 3, \ldots,$ or $2m$. But then by the induction hypothesis the remaining $2m-2$ numbers have exactly $(2m-3)!!$ pairings. Thus we have

$$(2m-1)!! = (2m-1)\cdot(2m-3)!!$$

which exactly matches the simple recurrence formula.

We now show the moments of a Gaussian are given in terms of the double factorial:

Lemma 1 (Gaussian moments)**.**

$$\mu_k := \int_{-\infty}^{\infty} x^k \frac{\mathrm{e}^{-x^2/2}}{\sqrt{2\pi}} \mathrm{d}x = \begin{cases} 0 & \textit{if } k \textit{ is odd} \\ (k-1)!! & \textit{if } k \textit{ is even} \end{cases}$$

Proof

We begin with a more formal explanation. The case of k odd follows from symmetry. For even k we use a proof by induction. The base case of $k=0$ follows using the classic trick of computing the integral of a Gaussian by turning it into a 2D integral, using polar coordinates $(r^2 = x^2+y^2)$ and the change of variables $t = r^2/2$:

$$\begin{aligned}\left(\int_{-\infty}^{\infty} \mathrm{e}^{-x^2/2}\mathrm{d}x\right)^2 &= \left(\int_{-\infty}^{\infty} \mathrm{e}^{-x^2/2}\mathrm{d}x\right)\left(\int_{-\infty}^{\infty} \mathrm{e}^{-y^2/2}\mathrm{d}y\right) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} \mathrm{e}^{-(x^2+y^2)/2}\mathrm{d}x\mathrm{d}y \\ &= \int_0^{\infty}\int_0^{2\pi} r\mathrm{e}^{-r^2/2}\mathrm{d}r\mathrm{d}\theta = 2\pi\int_0^{\infty}\mathrm{e}^{-t}\mathrm{d}t = 2\pi,\end{aligned}$$

hence

$$\mu_0 = \frac{\int_{-\infty}^{\infty} \mathrm{e}^{-x^2/2}\mathrm{d}x}{\sqrt{2\pi}} = 1.$$

For even $k > 0$ assume it is true for all smaller k. Then we can integrate-by-parts to deduce:

$$\begin{aligned}\sqrt{2\pi}\mu_k &= \int_{-\infty}^{\infty} x^k\mathrm{e}^{-x^2/2}\mathrm{d}x = \int_{-\infty}^{\infty} x^{k-1} \underbrace{x\mathrm{e}^{-x^2/2}}_{=-\frac{\mathrm{d}}{\mathrm{d}x}\mathrm{e}^{-x^2/2}} \mathrm{d}x \\ &= [-x^{k-1}\mathrm{e}^{-x^2/2}]_{-\infty}^{\infty} + (k-1)\int_{-\infty}^{\infty} x^{k-2}\mathrm{e}^{-x^2/2}\mathrm{d}x = \sqrt{2\pi}(k-1)\mu_{k-2}.\end{aligned}$$

I.e., $\mu_k = (k-1)\mu_{k-2}$, but this is exactly the formula that we showed $(k-1)!!$ satisfies above (if we write $k = 2m$).

■

I.3 Central Limit Theorem

We now want to show that the moments $\mathbb{E}[S_n^k]$ for

$$S_n := \frac{X_1 + \ldots + X_n}{\sqrt{n}}.$$

match those of the Gaussian, and hence are also given by the double factorial as $n \to \infty$. This implicitly tells us that the distribution converges to a Gaussian. We will use the *multinomial theorem*:

Theorem 1 (multinomial theorem)**.**

$$(X_1 + \cdots + X_n)^k = \sum_{k_1+\cdots+k_n=k} \frac{k!}{k_1! \cdots k_n!} X_1^{k_1} \cdots X_n^{k_n}.$$

Proof This is a generalisation of the binomial theorem, and can be proved by induction on k. The base case of $n = 1$ is trivial and the case $n = 2$ is equivalent to the binomial theorem. For $n \geq 2$ we write:

$$\begin{aligned}(X_1 + \cdots + X_{n-1} + X_n + X_{n+1})^k &= (X_1 + \cdots + X_{n-1} + (X_n + X_{n+1}))^k \\ &= \sum_{k_1+\cdots+k_{n-1}+\kappa=k} \frac{k!}{k_1! \cdots k_{n-1}!\kappa!} X_1^{k_1} \cdots X_{n-2}^{k_{n-2}} (X_n + X_{n+1})^\kappa \\ &= \sum_{k_1+\cdots+k_{n-1}+\kappa=k} \frac{k!}{k_1! \cdots k_{n-1}!\kappa!} X_1^{k_1} \cdots X_{n-2}^{k_{n-2}} \sum_{k_n+k_{n+1}=\kappa} \frac{\kappa!}{k_n!k_{n+1}!} X_n^{k_n} X_{n+1}^{k_{n+1}}\end{aligned}$$

where we have used the binomial theorem in the last step. But we now note we can combine the two sums as a single sum to deduce:

$$\begin{aligned}(X_1 + \cdots + X_{n-1} + X_n + X_{n+1})^k &= \sum_{k_1+\cdots+k_{n+1}=k} \frac{k!}{k_1! \cdots k_{n-1}!\kappa!} \frac{\kappa!}{k_n!k_{n+1}!} X_1^{k_1} \cdots X_{n+1}^{k_{n+1}} \\ &= \sum_{k_1+\cdots+k_{n+1}=k} \frac{k!}{k_1! \cdots k_{n-1}!k_n!k_{n+1}!} X_1^{k_1} \cdots X_{n+1}^{k_{n+1}}.\end{aligned}$$

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This basic result allows us to exapnd out the moments and deduce the main result:

Theorem 2 (CLT)**.** *Let $X_1, X_2, \ldots$ be iid variables with $\mathbb{E}[X_j] = 0$, $\mathbb{E}[X_j^2] = 1$, and all other moments are finite. Then for all k we have*

$$\mathbb{E}[S_n^k] \xrightarrow[n\to\infty]{} \mu_k.$$

Proof

We have from the multinomial theorem and independence:

$$\mathbb{E}[S_n^k] = n^{-k/2} \sum_{k_1+\cdots+k_n=k} \frac{k!}{k_1! \cdots k_n!} \mathbb{E}[X_1^{k_1}] \cdots \mathbb{E}[X_n^{k_n}].$$

We claim when $k = 2m$ that the sum is dominated by the terms where the k_j are either zero or 2, in particular, where there are precisely m 2s. In particular, we have to choose m indices to have $k_j = 2$ from a total of n, and there are exactly

$$\binom{n}{m} = \frac{n!}{m!(n-m)!}$$

of these. Noting that $\mathbb{E}[X_k^2] = \mathbb{E}[X_k^0]$, these terms each take a value of

$$n^{-m} \frac{(2m)!}{2^m} \mathbb{E}[X_1^{k_1}] \cdots \mathbb{E}[X_n^{k_n}] = \frac{(2m)!}{(2n)^m},$$

giving a total contribution of

$$\frac{n!(2m)!}{(2n)^m m!(n-m)!} \to \frac{(2m)!}{2^m m!} = (2m-1)!!$$

where we use the fact that, for m fixed,

$$\frac{n!}{(n-m)!n^m} = \frac{n(n-1)\cdots(n-m+1)}{n^m} = \frac{n^m + O(n^{m-1})}{n^m} \to 1$$

as $n \to \infty$.

We are just left with showing the total contribution from all other choices of $k_1+\cdots+k_n = k$ tend to zero. Note that the terms where any $k_j = 1$ contribute zero as $\mathbb{E}[X_j] = 0$. Showing the remaining terms don't contribute as $n \to \infty$ (and that in the odd case no terms contribute) is a bit technical so we omit the details.

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Remark Note the assumption of finite moments is essential. A classic counter example is if X_k have a heavy-tail distribution like the Cauchy distribution:

$$\frac{1}{\pi}\frac{1}{x^2+1}.$$

Not even the first moment exists, that is, this distribution has no mean! (Symmetry may make one think that the mean should be zero, but if we average samples it never converges.) If we take iid samples X_k of the Cauchy distribution we know that

$$\frac{X_1 + \cdots + X_n}{n}$$

is the same Cauchy distribution. (Note the division by n, not $\sqrt{n}$, which differs from the CLT above.)

I.4 Wigner ensembles and global statistics

To discuss universality in the setting of random matrices we focus on symmetric matrices with iid entries, i.e., the *Wigner ensembles*:

Definition 2 (Wigner)**.** A random matrix $A_n \in \mathbb{R}^{n\times n}$ is a Wigner matrix if:

1. The entries a_{kj} for $j \geq k$ are all iid with mean 0, variance 1, and all moments are finite.
2. The matrix A_n is symmetric: $a_{kj} = a_{jk}$.

We are interested in the global distribution of the eigenvalues. A Wigner ensemble $A_n \in \mathbb{R}^{n\times n}$ has n eigenvalues $\lambda_1, \ldots, \lambda_n$ which can be viewed as n random variables in their own right. Their distribution can be described in terms of the moments:

$$\sigma_{nk} := \mathbb{E}[\lambda_1^k + \ldots + \lambda_n^k].$$

We want to show that as $n \to \infty$ that these go to the moments of the semicircle. But before that we observe that we can rewrite these moments in terms of A_n itself via:

$$\sigma_{nk} = \mathbf{E}A_n^k = \mathbb{E}[\operatorname{Tr} A_n^k]$$

for the trace $\operatorname{Tr} A_n = a_{11} + \cdots + a_{nn}$. This follows from the following:

Proposition 1 (Cyclic property of trace). *For any $A \in \mathbb{R}^{n\times m}$ and $B \in \mathbb{R}^{m\times n}$ we have*

$$\operatorname{Tr} AB = \operatorname{Tr} BA$$

Proof

We show this by direct inspection:

$$\begin{aligned}\operatorname{Tr} AB &= \sum_{k=1}^n \boldsymbol{e}_k^\top AB\boldsymbol{e}_k = \sum_{k=1}^n \begin{bmatrix} a_{k1} & \cdots & a_{km}\end{bmatrix}\begin{bmatrix} b_{1k} \\ \vdots \\ b_{m1}\end{bmatrix} = \sum_{k=1}^n\sum_{j=1}^m a_{kj}b_{jk} \\ &= \sum_{j=1}^m\sum_{k=1}^n b_{jk}a_{kj} = \sum_{j=1}^m \begin{bmatrix} b_{j1} & \cdots & b_{jn}\end{bmatrix}\begin{bmatrix} a_{j1} \\ \vdots \\ a_{jn}\end{bmatrix} = \sum_{j=1}^m \boldsymbol{e}_j^\top BA\boldsymbol{e}_j = \operatorname{Tr} BA.\end{aligned}$$

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Proposition 2 (Traces, powers and eigenvalues). *For any symmetric $A \in \mathbb{R}^{n\times n}$ with eigenvalues $\lambda_1, \ldots, \lambda_n$ we have*

$$\operatorname{Tr} A^k = \lambda_1^k + \ldots + \lambda_n^k.$$

Proof

We know that a symmetric matrix matrix is *diagonalisable*:

$$A = V \underbrace{\begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n\end{bmatrix}}_{\Lambda} V^{-1}.$$

Therefore

$$A^k = V\Lambda^k V^{-1}.$$

The proposition follows since the trace is equal to the sum of the eigenvalues, which follows from the cyclic property:

$$\operatorname{Tr} A^k = \operatorname{Tr} V(\Lambda^k V^{-1}) = \operatorname{Tr}(\Lambda^k V^{-1})V = \operatorname{Tr}\Lambda^k = \lambda_1^k + \ldots + \lambda_n^k.$$

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Thus we have a plan of attack: (1) compute the limit of $\sigma_{nk} = \mathbf{E}A_n^k$ as $n \to \infty$ and (2) show this matches the moments of the semicircle distribution.

I.5 Moments of the semicircle distribution

We now establish the moments of the semicircle distribution, using a combinatorial construction with a similar flavour (but not the same!) as the normal distribution. In particular they will be given in terms of the Catalan numbers (instead of the double factorial):

Definition 3 (Catalan numbers).

$$C_n := \frac{1}{n+1}\binom{2n}{n} = \frac{(2n)!}{(n+1)!n!}.$$

We can compute the first few:

$$C_0 = C_1 = 1, C_2 = \frac{4!}{3!2!} = 2, C_3 = \frac{6!}{4!3!} = 5,$$

and indeed these match are prediction for the even moments.

A key property we will use is that Catalan numbers satisfy a simple recurrence:

Lemma 2 (Catalan recurrence)**.**

$$C_{n+1} = \sum_{k=0}^{n} C_k C_{n-k}.$$

Proof

This proof is a quite involved generating function calculation. Going through it could be the basis of a poster project.

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Catalan numbers are equivalent to a wide range of different combinatorial objects including (1) non-crossing partitions of n, (2) the number of Dyck words of length $2n$, (3) the number of full binary trees with $n+1$ leaves, etc.

Here we are interested in the fact that they tell us exactly how many ways to partition $\{1, \ldots, 2n\}$ into pairs (like the Gaussian moments), but now with the restriction that the pairs are *non-crossing*: for $k \neq j$ we never have $x_k < x_j < y_k < y_j$. For example, when $n = 2$ we have $C_2 = 4!/(3!2!) = 2$ non-crossing pairings:

$$\{1, 2\}, \{3, 4\} \qquad \{1, 4\}, \{2, 3\},$$

which excludes the pairing $\{1, 3\}, \{2, 4\}$ since $1 < 3 < 4 < 2$. (Note in the second non-crossing pair the second pair lies completely within the first pair, so they don't cross.) When $n = 3$ we have $C_3 = 6!/(4!3!) = 5$ non-crossing pairings:

$$\begin{array}{lll} \{1, 2\}, \{3, 4\}, \{5, 6\} & \{1, 2\}, \{3, 6\}, \{4, 5\} & \\ \{1, 4\}, \{2, 3\}, \{5, 6\} & \{1, 6\}, \{2, 3\}, \{4, 5\} & \{1, 6\}, \{2, 5\}, \{3, 4\}. \end{array}$$

To see the number of such partitions are given by Catalan numbers we will do a proof by induction, noting that the case of $C_1 = 1$ is immediate (there is only one way to pair two numbers). Observe the following:

1. If 1 is paired with 2, there are $2(n-1)$ remaining numbers to pair.
2. The number 1 cannot be paired with 3 as then there is no possibility of pairing 2 with another number without crossing with $\{1, 3\}$.
3. If 1 is paired with 4 then $\{2, 3\}$ must be a pair and there are $2(n-2)$ remaining numbers to pair.
4. Th number 1 cannot be paired with 5 since we cannot pair $\{2, 3, 4\}$ without having a left over. By the same logic, 1 cannot be paired with any odd number.
5. If 1 is paired with an even number $2k$ we have to pair the $2(k-1)$ numbers $\{2, \ldots, 2k-1\}$ and the $2(n-k)$ numbers $\{2k+1, \ldots, 2n\}$.

Putting all this together we have, assuming C_k gives the number of such pairings of $2k$ for $k < n$,

$$\underbrace{C_{n-1}}_{\text{when } \{1,2\} \text{ is a pair}} + \underbrace{C_{n-2}}_{\text{when } \{1,4\} \text{ is a pair}} + \underbrace{C_2 C_{n-3}}_{\text{when } \{1,6\} \text{ is a pair}} + \underbrace{C_3 C_{n-4}}_{\text{when } \{1,8\} \text{ is a pair}} + \cdots + \underbrace{C_{n-1}}_{\text{when } \{1,2k\} \text{ is a pair}} .$$

using the fact that $C_0 = C_1 = 1$ this reduces to

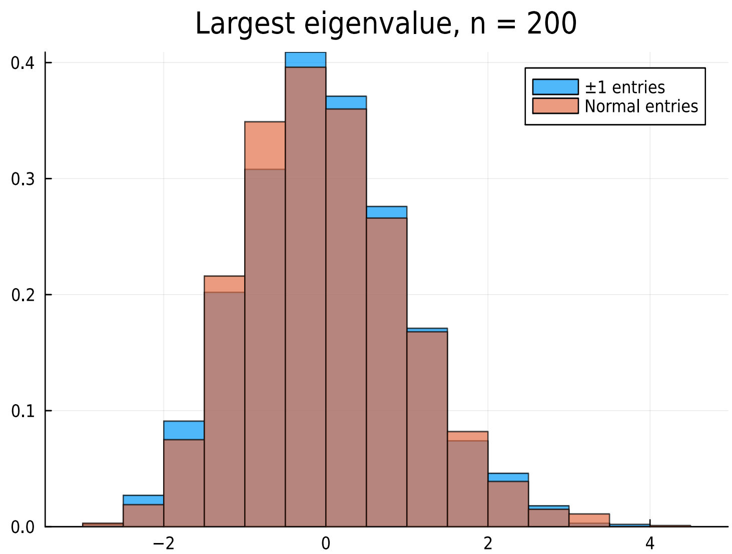
$$\sum_{k=0}^{n-1} C_k C_{n-k-1} = C_n$$

by the above recurrence, and hence the result is true by induction.

I.6 Proof of the semicircle law

I.7 Local eigenvalue statistics

One can also ask about the statistics of the largest eigenvalue. Here we plot a Histogram of the largest eigenvalue:



I.8 Poster projects

The possibilities for projects related to universality are endless and I encourage you to think about an original idea. Here are some example projects to help guide you:

1. Hermitian random matrices also satisfy universality properties. How do histograms of their eigenvalue statistics (both global and local) compare to the symmetric case?

2. A *Wishart ensemble* is a random symmetric matrix $XX^\top$ where $X \in \mathbb{R}^{m\times n}$ is a rectangular matrix with random entries. It also exhibits universality but depending on the ratio $\alpha = m/n$. Can you explore how the global and local statistics behave? Do they match the Wigner ensemble case?

3. The spacing of zeros of the Riemann Zeta function on the critical line match the universality. to higher accuracy?

4. What happens when the 4-moment condition breaks down? I.e. what are eigenvalue statistics for matrices with heavytailed entries?

5. One can consider matrices associated with random networks, a la as discussed in Introduction to Applied Mathematics. How connected does a random network need to be to exhibit universality?

Chapter II

Free Probability

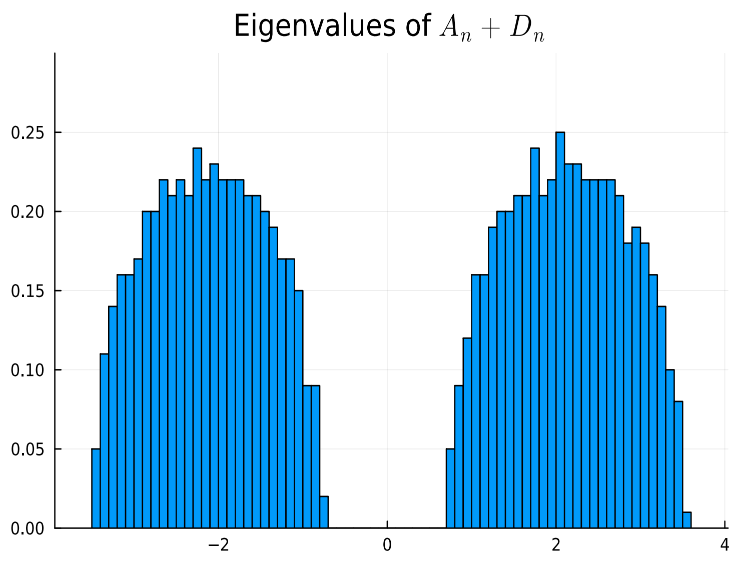
We now switch our attention and ask: given two (symmetric) random matrices $A, B \in \mathbb{R}^{n\times n}$ and B, what are the distribution of the eigenvalues of $A+B$? The rather remarkable fact is that, as $n \to \infty$, if we know the global distribution of A and B, in the form of knowing the moments $\mathbf{E}A^k$ and $\mathbf{E}B^k$, under very broad conditions we can deduce the global distribution of $A+B$, in the form of knowing the moments $\mathbf{E}(A+B)^k$, where again we use

$$\mathbf{E}f(A) := \mathbb{E}\mathrm{Tr}\, f(A).$$

The key observation is that A and B, which do not in general commute ($AB \neq BA$), follow the rules of *freely independent* non-commutative random variables. Thus the global statistics of $A+B$ can be thought of as a *free convolution*, a non-commutative analogue of convolution.

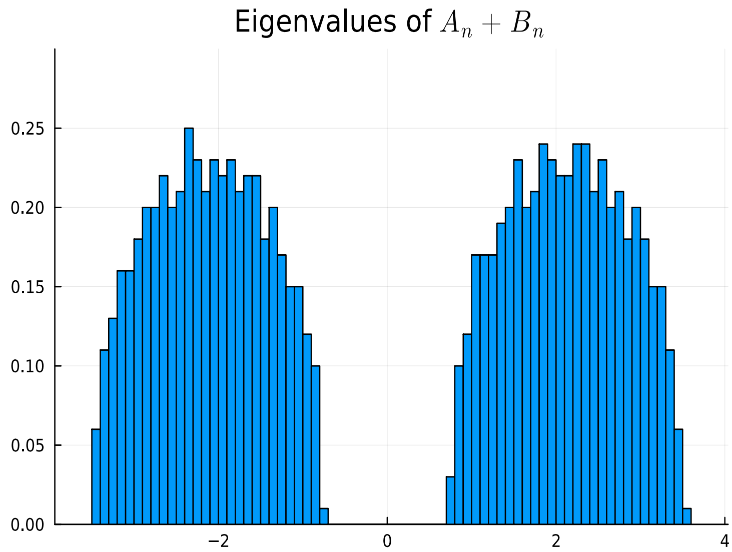
II.1 Experimental observation of free convolutions

We begin with an experimental observation of free convolution. Consider again a symmetric matrix $A_n \in \mathbb{R}^{n\times n}$ whose entries on and above the diagonal are iid random variables taking values ± 1 with equal probability. We saw in the previous chapter the global statistics are given by the semicircle. Now consider adding a diagonal matrix $D_n \in \mathbb{R}^{n\times n}$ with ± 2 on the diagonal with equal probability: what are the global statistics of $A_n + D_n$? We can explore this question numerically:



We see that we get two semicircle-like distributions, surrounding ± 2.

What's remarkable is this distribution only depends on the eigenvalues of A_n and D_n. For example, consider the case where we same randomly $Q_n \in O(n)$, i.e., a random orthogonal matrix and define $B_n := Q_n D_n Q_n^\top$. This has exactly the same eigenvalues as D_n itself. And when we add $A_n + B_n$ we get the exact same global statistics:



This is a numerical demonstration of the fact that we can, under broad conditions, deduce the global eigenvalue statistics of $A_n + B_n$ from those of A_n and B_n, and largely ignore the eigenvectors. To prove this we will again use the Method of Moments.

II.2 Free random variables

We put aside viewing A and B as random matrices and instead think of them more abstractly as *free random variables*, which are non-commutative random variables which satisfy the following:

Definition 4 (free random variables)**.** A and B are *free random variables* equipped with an expectation $\mathbf{E}$ if for all polynomials $f_1, \ldots, f_n$ and $g_1, \ldots, g_n$ such that $\mathbf{E} f_k(A) = \mathbf{E} g_k(B) = 0$ we have

$$\mathbf{E} f_1(A) g_1(B) \ldots f_n(A) g_n(A) = 0,$$

where the identity element satisfies $\mathbf{E} I = 1$.

We claim (without proof) that as $n \to \infty$ that $\mathbf{E} := \mathbb{E}\mathrm{Tr}$ behaves exactly like this for many families of random matrices.

Note that we assume linearity: for constants c, d we have

$$\mathbf{E}(cA + dB) = c\mathbf{E}A + d\mathbf{E}B.$$

It follows that (treating $\mathbf{E}A^k$ as a constant)

$$\mathbf{E}(A^k - (\mathbf{E}A^k)I) = \mathbf{E}A^k - \mathbf{E}A^k\mathbf{E}I = 0.$$

This simple observation guarantees that if we know the moments $\mathbf{E}A^k$ and $\mathbf{E}B^k$ we can deduce the moments $\mathbf{E}A^k B^j$. To see this, we first consider some of the low order cases.

First, we note that by subtracting off the free expectation of moments of A and B and treating $\mathbf{E}[A^k], \mathbf{E}[B^j]$ as constants we deduce:

$$\begin{aligned} 0 &= \mathbf{E}[(A^k - \mathbf{E}[A^k]I)(B^j - \mathbf{E}[B^j]I)] = \mathbf{E}[A^k B^j] - \mathbf{E}A^k \mathbf{E}B^j \qquad \Rightarrow \\ \mathbf{E}[A^k B^j] &= \mathbf{E}A^k \mathbf{E}B^j \end{aligned}$$

which is identical to classical random variables. This apparent consistency with classic random variables works for one mixed moment:

$$\begin{aligned} 0 &= \mathbf{E}[(A - \mathbf{E}[A]I)(B - \mathbf{E}[B]I)(A - \mathbf{E}[A]I)] \\ &= \mathbf{E}[ABA] - \mathbf{E}[A]\mathbf{E}[AB] - \mathbf{E}[B]\mathbf{E}[A^2] - \mathbf{E}[A]\mathbf{E}[BA] + 2\mathbf{E}[A]^2\mathbf{E}[B] \\ &= \mathbf{E}[ABA] - \mathbf{E}[B]\mathbf{E}[A^2] \qquad \Rightarrow \\ \mathbf{E}[ABA] &= \mathbf{E}[A^2]\mathbf{E}[B] \end{aligned}$$

where we use $\mathbf{E}[AB] = \mathbf{E}[A]\mathbf{E}[B]$. But that's where the equivalence ends. In particular, consider:

$$\begin{aligned} 0 &= \mathbf{E}[(A - \mathbf{E}[A]I)(B - \mathbf{E}[B]I)(A - \mathbf{E}[A]I)(B - \mathbf{E}[B]I)] \\ &= \mathbf{E}[ABAB] - \underbrace{\mathbf{E}[ABA]}_{=\mathbf{E}[A^2]\mathbf{E}[B]} \mathbf{E}[B] - \mathbf{E}[A] \underbrace{\mathbf{E}[AB^2]}_{=\mathbf{E}[A]\mathbf{E}[B^2]} + \underbrace{\mathbf{E}[AB]}_{=\mathbf{E}[A]\mathbf{E}[B]} \mathbf{E}[A]\mathbf{E}[B] \\ &\quad - \underbrace{\mathbf{E}[A^2B]}_{=\mathbf{E}[A^2]\mathbf{E}[B]} \mathbf{E}[B] + \mathbf{E}[A^2]\mathbf{E}[B]^2 + \underbrace{\mathbf{E}[AB]}_{=\mathbf{E}[A]\mathbf{E}[B]} \mathbf{E}[A]\mathbf{E}[B] - \mathbf{E}[A]^2\mathbf{E}[B]^2 \\ &\quad - \mathbf{E}[A] \underbrace{\mathbf{E}[BAB]}_{=\mathbf{E}[A]\mathbf{E}[B^2]} + \mathbf{E}[A]\mathbf{E}[B] \underbrace{\mathbf{E}[BA]}_{=\mathbf{E}[A]\mathbf{E}[B]} + \mathbf{E}[A]^2\mathbf{E}[B^2] - \mathbf{E}[A]^2\mathbf{E}[B]^2 \\ &\quad + \mathbf{E}[A]\mathbf{E}[B] \underbrace{\mathbf{E}[AB]}_{=\mathbf{E}[A]\mathbf{E}[B]} - \mathbf{E}[A]^2\mathbf{E}[B]^2 - \mathbf{E}[A]^2\mathbf{E}[B]^2 + \mathbf{E}[A]^2\mathbf{E}[B]^2 \\ &= \mathbf{E}[ABAB] - \mathbf{E}[A^2]\mathbf{E}[B]^2 - \mathbf{E}[A]^2\mathbf{E}[B^2] + \mathbf{E}[A]^2\mathbf{E}[B]^2 \qquad \Rightarrow \\ \mathbf{E}[ABAB] &= \mathbf{E}[A^2]\mathbf{E}[B]^2 + \mathbf{E}[A]^2\mathbf{E}[B^2] - \mathbf{E}[A]^2\mathbf{E}[B]^2 \end{aligned}$$

which differs from the classical (commutative) case where $\mathbf{E}[ABAB] = \mathbf{E}[A^2]\mathbf{E}[B^2]$. Nevertheless, we can see the central theme emerging: $\mathbf{E}[ABAB]$ is still given in terms of a function of the input moments $\mathbf{E}[A^k]$ and $\mathbf{E}[B^j]$.

II.3 Free convolution

II.4 Free Central Limit Theorem

II.5 Poster projects

Here are some ideas for poster projects related to free probability:

1. Understanding all free probability operations including multiplication and truncation.
2. What are the heavytail analogues of the Semicircle Law?
3. Do additions and multiplications of random matrices still exhibit universality in local statistics?

Chapter III

Randomised Linear Algebra

Our last topic is *randomised linear algebra*: how can we incorporate random matrices into developing better algorithms for linear algebra?

Martinsson & Tropp 2020

III.1 Randomised trace estimation

We begin with a simple question: can we compute the trace of a matrix without computing all the entries on the diagonal? Often one has a matrix $A \in \mathbb{R}^{n\times n}$ that we know how to compute matrix-vector $A\boldsymbol{x}$, lets say in $O(n^2)$ operations, but we do not have direct access to the entries of the matrix a_{kj}. Of course, we can deduce the entries of a matrix via $a_{kj} = \boldsymbol{e}_k^\top (A\boldsymbol{e}_j)$, and hence we can compute the trace via:

$$\operatorname{Tr} A = \sum_{k=1}^n a_{kk} = \sum_{k=1}^n \boldsymbol{e}_k^\top A\boldsymbol{e}_k,$$

but this takes $O(n^3)$ operations.

By the cyclic property of trace we have

$$\operatorname{Tr} A = \operatorname{Tr} Q^\top A Q = \sum_{k=1}^n \boldsymbol{q}_k^\top A\boldsymbol{q}_j$$

for any orthogonal matrix Q. Let's do an experiment where we take a fixed matrix A:

```
0.9303525062438385
```

III.2 Least squares

We begin with an introduction to least squares. For a rectangular matrix $A \in \mathbb{R}^{m\times n}$ with $m > n$ and a right-hand side $\boldsymbol{b} \in \mathbb{R}^m$ we cannot solve $A\boldsymbol{c} = \boldsymbol{b}$ unless $\boldsymbol{b}$ happens to be in the column span of A. However, we can instead find $\boldsymbol{c} \in \mathbb{R}^n$ that satisfies $A\boldsymbol{c} \approx \boldsymbol{b}$, in the sense that

$$\|A\boldsymbol{c} - \boldsymbol{b}\|$$

is minimised. This is called the *least squares* solution. Minimising a norm is equivalent to minimising its square, and expanding out we see that it is equivalent to a quadratic minimisation problem:

$$\|A\boldsymbol{c}-\boldsymbol{b}\|^2 = (A\boldsymbol{c}-\boldsymbol{b})^\top (A\boldsymbol{c}-\boldsymbol{b}) = \boldsymbol{c}^\top A^\top A\boldsymbol{c} - 2\boldsymbol{c}^\top A^\top \boldsymbol{b} + \boldsymbol{b}^\top \boldsymbol{b}.$$

In the scalar case ($n = 1$ so that $\boldsymbol{c}$ is essentially a constant) we know we can solve this by differentiation with respect to $\boldsymbol{c}$, and we arrive at a linear system

$$A^\top A\boldsymbol{c} = A^\top \boldsymbol{b}$$

This generalises (after some linear algebra) and in general we have that

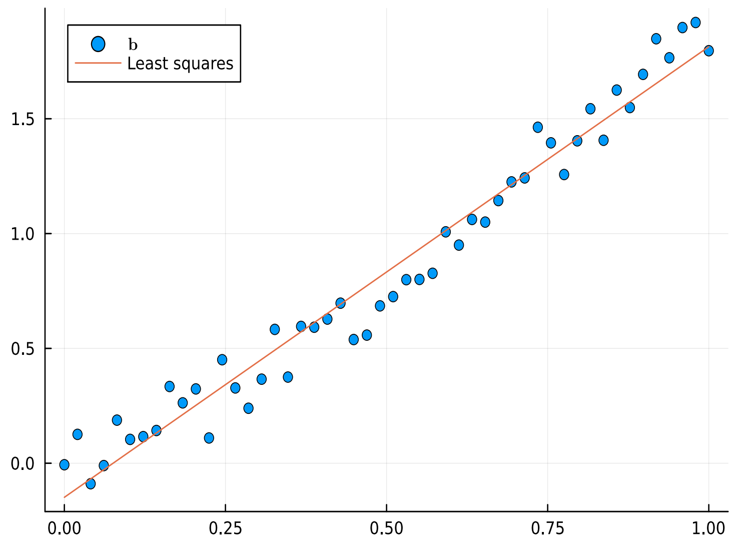
$$\boldsymbol{c} = (A^\top A)^{-1} A^\top \boldsymbol{b}.$$

A classic application of least squares is *polynomial regression*. In the case of linear regression, we want to approximation data $\{b_j\}$ on a grid of points $\{x_j\}$ by a linear function $c_0 + c_1 x$. In other words, we want to solve

$$\underbrace{\begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} c_0 \\ c_1 \end{bmatrix}}_{\boldsymbol{c}} \approx \underbrace{\begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}}_{\boldsymbol{b}}$$

by using the least squares solution.

Here is a depiction of $c_0 + c_1 x$, which is the least squares fit at the (noisy) data at $n = 50$ points:

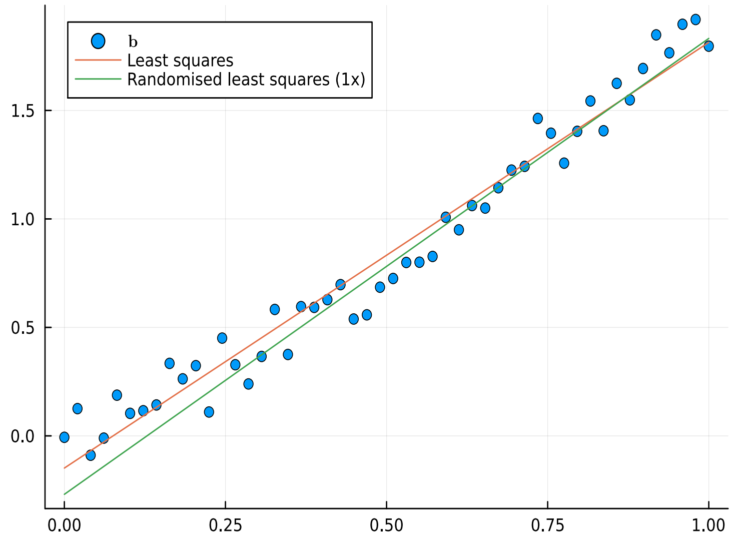


III.3 Randomised least squares

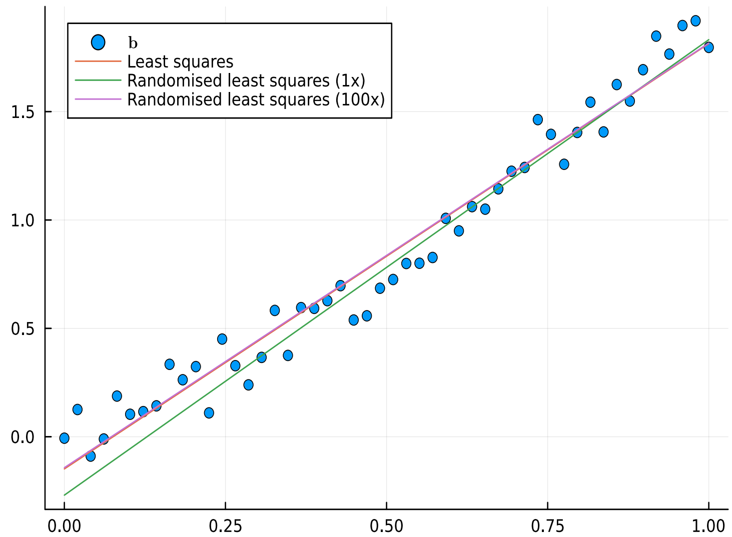
In the above data there is a lot of correlation between nearby points, hence, using every single point is unnecessary. An alternative is to *sketch* the problem: use a sketch matrix $P \in \mathbb{R}^{r\times m}$ where $r \ll m$ and solve

$$PA\boldsymbol{c} \approx P\boldsymbol{b}$$

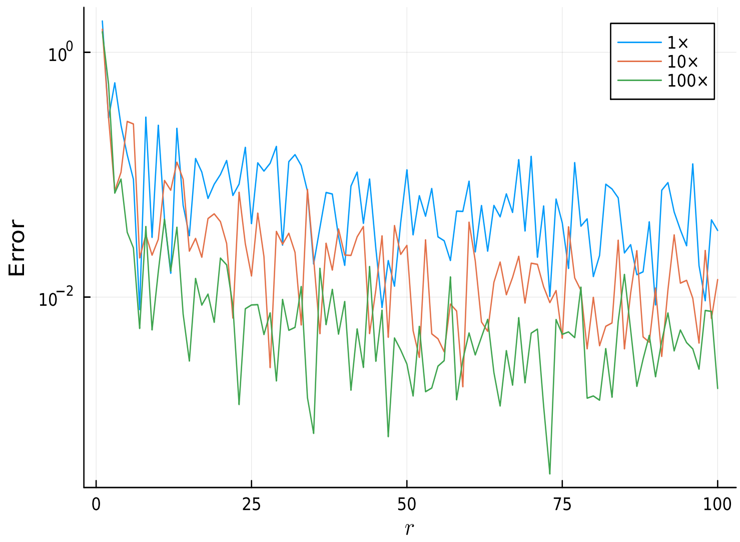
using least squares. Here we see a simple example where we use $P \in \mathbb{R}^{5\times 50}$ with random entries:



This is a single sample and, to the eye, it performs equally well as least squares. Going further, we see that averaging over multiple trials is almost indistiguishable from the true least squares approximation:



Now this particular example is a bit silly: its not at all clear that solving 100 least squares systems of size 5×2 is preferred to a single least squares system of size 50×2. But we can consider a much more extreme example of $n = 10,000$. Here we compare the average over samples for varying r:



That is to say, we can achieve accuracy to within about 1% by solving 100 least squares problems involving 10×2 matrices.

III.4 Poster projects

1. How does the approximation property depend on the distribution of the sketch matrix?
2. What other algorithms are amenable to randomisation (low rank approximation, norm estimation, multiplication)?
3. Can averaging over multiple sketch matrices be used to improve accuracy while preserving speed advantage?
4. What applications are there to randomised linear algebra such as regression?